

International Institute of Information Technology, Bangalore
CSE 511 Algorithms: Practice Problem

Description: There are several squares being dropped onto the X-axis of a 2D plane. You are given a 2D integer array `positions` where `positions[i] = [left_i, sideLength_i]` represents the i -th square with a side length of `sideLength_i` that is dropped with its left edge aligned with X-coordinate `left_i`.

Each square is dropped one at a time from a height above any landed squares. It then falls downward (negative Y direction) until it either lands **on the top side of another square** or **on the X-axis**. A square brushing the left/right side of another square does not count as landing on it. Once it lands, it freezes in place and cannot be moved.

After each square is dropped, you must record the **height of the current tallest stack of squares**. Return an integer array `ans` where `ans[i]` represents the height described above after dropping the i -th square[reference:0].

Input:

- The first line contains an integer n , the number of squares.
- The next n lines each contain two integers: `left_i` and `sideLength_i`.

Output: Print the array `ans` of length n , where `ans[i]` is the tallest stack height after dropping the first $i + 1$ squares.

Example 1:

Input :

```
3
1 2
2 3
6 1
```

Output:

```
[2, 5, 5]
```

Explanation:

- After the first drop (`[1, 2]`), the tallest stack is square 1 with a height of 2.
- After the second drop (`[2, 3]`), the second square lands on top of the first square (they overlap in the range `[2, 3]`). The tallest stack is squares 1 and 2 with a height of $2 + 3 = 5$.
- After the third drop (`[6, 1]`), the third square lands on the X-axis. The tallest stack is still squares 1 and 2 with a height of 5.

Example 2:

Input:

2

100 100

200 100

Output:

[100, 100]

Explanation: After the first drop, the tallest stack is square 1 with a height of 100. After the second drop, the tallest stack is either square 1 or square 2, both with heights of 100. Note that square 2 only brushes the right side of square 1, which does not count as landing on it[reference:1].

Constraints:

- $1 \leq \text{positions.length} \leq 1000$
- $1 \leq \text{left}_i \leq 10^8$
- $1 \leq \text{sideLength}_i \leq 10^6$